

Linear colorings of the binomial random graph

Austin Eide & Paweł Prałat

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6 March 2024

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Linear
colorings of
the binomial
random graph

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Centered &
linear colorings

Coloring
random
graphs

Cores,
expansion, and
long paths

Conclusions

For connected G , define **elimination tree** $T(G)$ recursively:

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Algorithm (Build E-tree)

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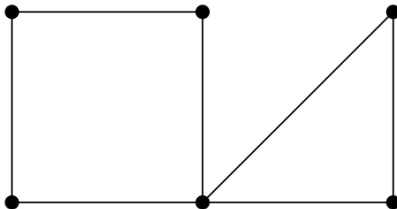
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The **tree-depth**, $\text{td}(G)$, is the minimum depth of an elimination tree for G .

Elimination tree

Example



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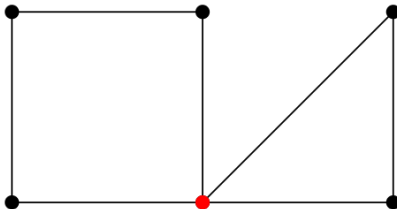
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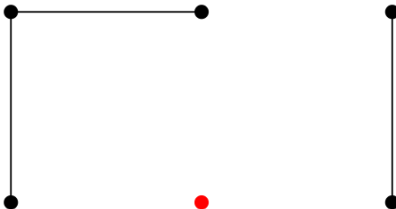
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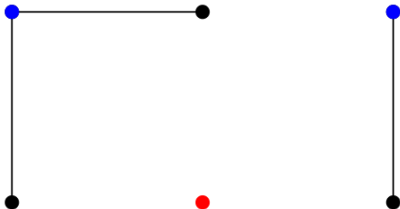
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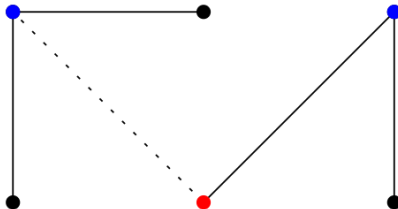
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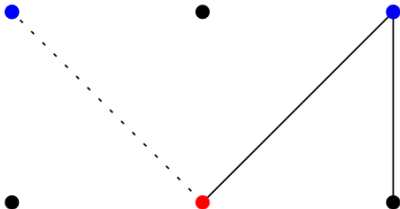
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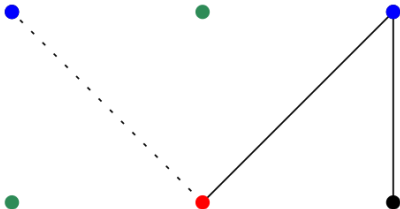
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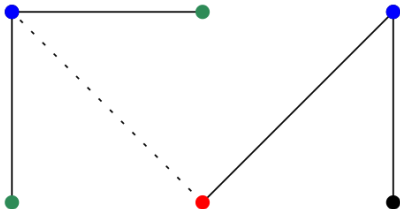
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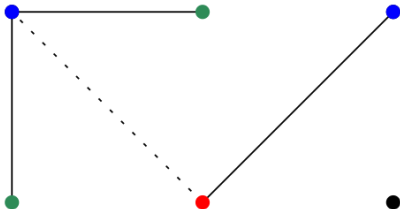
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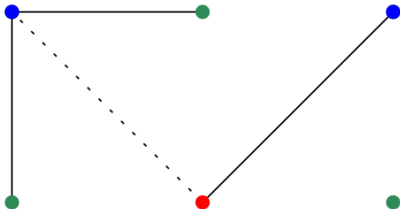
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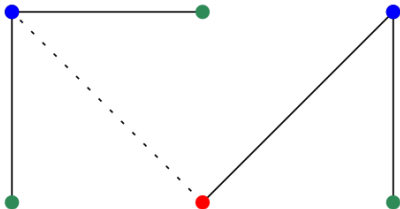
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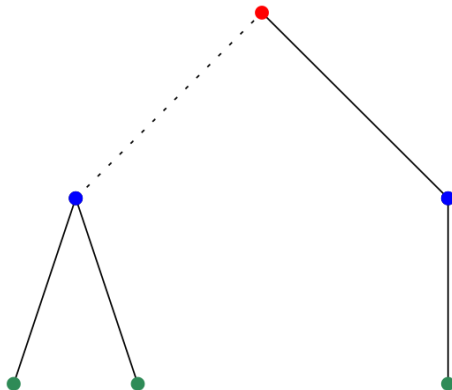
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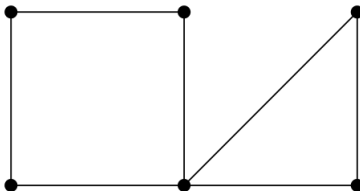
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is at most 3.

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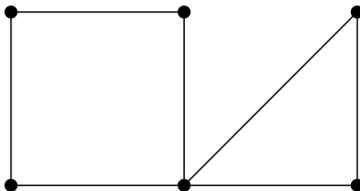
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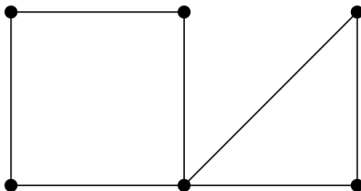
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Claim

Coloring “by level” gives a proper coloring, so $\chi(G) \leq td(G)$.

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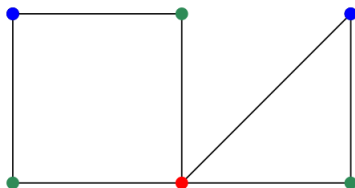
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Centered colorings $\iff$ *elimination trees*.

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Centered colorings $\iff$ elimination trees. For any G , $td(G) = \chi_{\text{cen}}(G)$.

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Conversely, easy algorithm converts centered k -coloring to depth- k e-tree. □

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- $\text{td}(K_n) = n$
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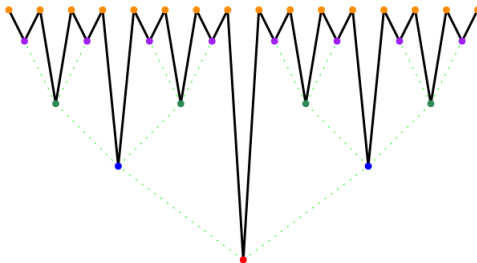
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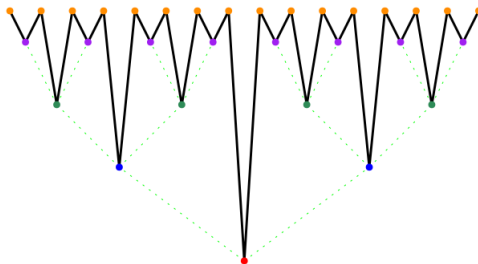
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- $\text{td}(G) \geq \log_2(\lceil \text{diam}(G) + 1 \rceil)$

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Application

Hard problems solved efficiently on graphs with “bounded tree-depth decompositions” (Ossona de Mendez, Nešetřil).

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Theorem (Pothen '88)

Shortest elimination tree is NP-complete.

Applications and complexity

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Efficient approximation algorithms?

Definition (Linear coloring, Kun, O'Brien, Pilipczuk, Sullivan '21)

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Proposition

$$\chi(G) \leq \chi_{\text{lin}}(G) \leq \chi_{\text{cen}}(G).$$

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Centered coloring $\implies$ linear coloring

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Proof.

Centered coloring $\implies$ linear coloring $\implies$ proper coloring. □

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On the other hand,

Theorem (KOPS '21, Bose, Dujmović, Houdrouge, Javarsineh, Morin '22)

$$\chi_{cen}(G) \leq \text{poly}(\chi_{lin}(G)).$$

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Proposition (KOPS '21)

For any $\epsilon > 0$, $\exists$ a G with $\chi_{cen}(G) > (2 - \epsilon)\chi_{lin}(G)$.

The binomial random graph $\mathcal{G}(n, p)$

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Definition

$\mathcal{G}(n, p)$ is the probability space on graphs on vertex set $[n]$ with each edge $\{i, j\}$ present ind. with probability p .

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With high probability

A sequence of events $\{\mathcal{A}_n\}$ in $\mathcal{G}(n, p)$ holds *with high probability* (w.h.p.) if $\Pr(\mathcal{A}_n) \xrightarrow{n \rightarrow \infty} 1$.

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Theorem (E. and Prałat '23)

If $\omega = np \rightarrow \infty$, then w.h.p.

$$\chi_{\text{lin}}(\mathcal{G}(n, p)) \geq n - O(n/\sqrt{\omega}).$$

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Theorem (Perarnau & Serra '14)

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Proof sketch

"Pairings" (Alon, McDiarmid, Reed)

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Suppose we color with $(1 - x)n$ colors:



$$(1 - x)n$$

Proof sketch

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Make all classes even by removing at most $(1 - x)n$ vertices:



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Make all classes even by removing at most $(1-x)n$ vertices:



(Arbitrarily) make pairings within remaining classes:



(There are at least $xn/2$ pairs.)

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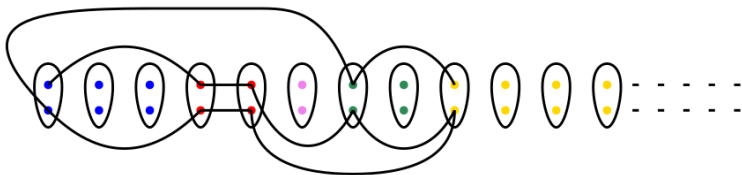
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A path that visits each pair exactly twice or not at all is **bad**.



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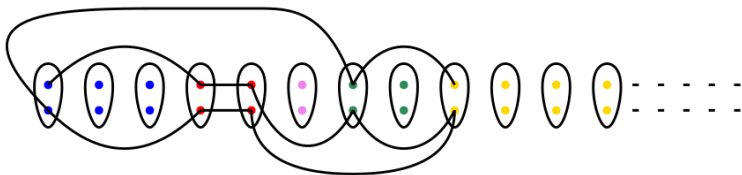
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To show the coloring is *not* linear, show the existence of a **bad path** on the pairing.

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"Pairings" (Alon, McDiarmid, Reed)

Lemma

Let $x = \frac{C}{\sqrt{\omega}}$. W.h.p., for any $S \subseteq [n]$ with $|S| \geq xn$ and any pairing $\mathcal{P}$ of vertices in S ...

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$\mathcal{G}(n, p)$ contains a **bad path** with respect to $S, \mathcal{P}$.

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Proof of main theorem.

$(1 - x)n$ -coloring $\implies$ pairing on $\geq xn$ vertices $\implies$ bad path by lemma. □

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Definition

Refer to an even set S and a pairing $\mathcal{P}$ of S collectively as a **set-pairing** $(S, \mathcal{P})$.

Long paths or short?

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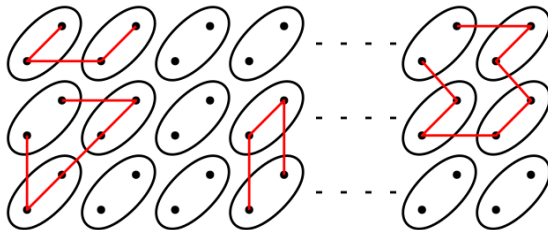
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Conclusions

When p is large, expect to see short bad paths...



Long paths or short?

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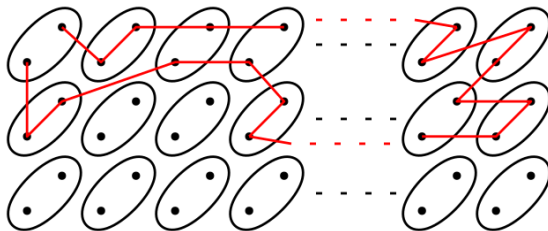
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...when p is small, first moment argument implies any bad path must be long.



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Roadblock:

Many sets S will have vertices of low or even zero degree in their induced subgraphs, even for $p = \Omega(1)$.

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Claim

W.h.p., for any $(S, \mathcal{P})$ with $|S| \geq xn$, by removing at most $|S|/4$ low-degree pairs we can find sub-pairing $(S', \mathcal{P}')$ so that S' induces a subgraph of min. degree $\geq |S|p/3 = \Omega(\sqrt{w})$

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Like a k -core, but respects the pairing $\mathcal{P}$.

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Algorithm (k -core)

Let $S_0 = S$, $\mathcal{P}_0 = \mathcal{P}$.

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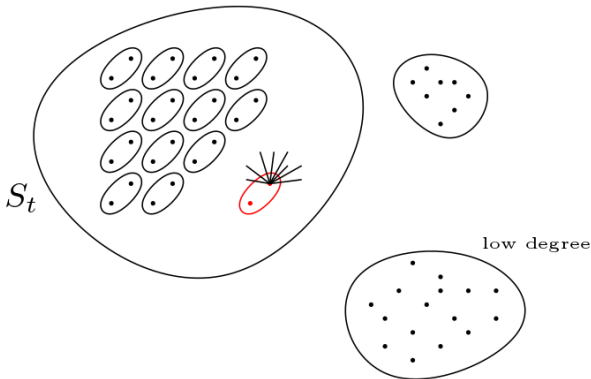
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Algorithm (k -core)

Let $S_0 = S$, $\mathcal{P}_0 = \mathcal{P}$. While S_t has a vertex of small degree in $G[S_t]$, remove its pair. Update S_{t+1} and $\mathcal{P}_{t+1}$.

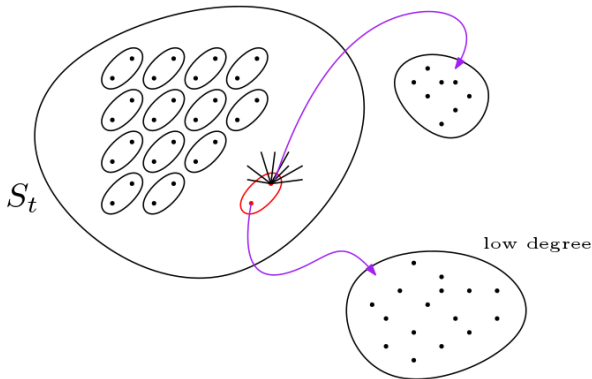
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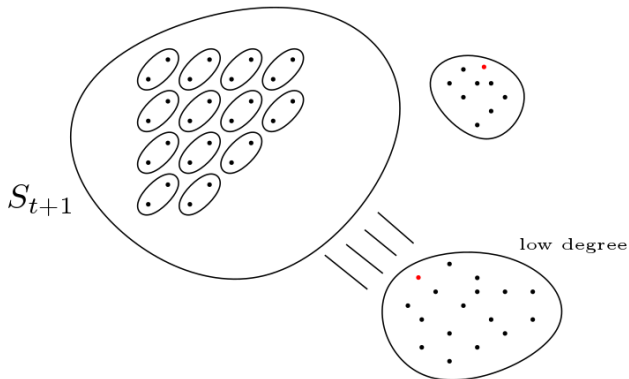


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Analysis

If the process runs for too long, we find large disjoint sets which induce a very sparse cut. But no such sets w.h.p.



Cores are good expanders

Cores are large and have high min. degree. Even better...

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- ② *Satisfies $|N_H(X)| \geq 2|X|$ for every $X \subset V(H)$ with $|X| \leq |V(H)|/45$.*

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Definition

G an $(r, 2)$ -expander if $|N(X)| \geq 2|X|$ for every $X \subset V(G)$ with $|X| \leq r$.

Pósa rotations

Standard technique for long paths in random graphs

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Two-round exposure

Expose $\mathcal{G}(n, p)$ in two rounds. Let $p = p_1 + p_2 - p_1 p_2$

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Expose $\mathcal{G}(n, p)$ in two rounds. Let $p = p_1 + p_2 - p_1 p_2$

- 1 Reveal copy of $\mathcal{G}(n, p_1)$.
- 2 Independently reveal copy of $\mathcal{G}(n, p_2)$

Then $\mathcal{G}(n, p) = \mathcal{G}(n, p_1) \cup \mathcal{G}(n, p_2)$.

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Take $p_1 = \frac{p}{2}$, $p_2 = \frac{p}{2} + \epsilon$.

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Take $p_1 = \frac{p}{2}$, $p_2 = \frac{p}{2} + \epsilon$.

Lemma

After Round 1, w.h.p. every $(S, \mathcal{P})$ with $|S| \geq xn$ has a large sub-pairing $(S', \mathcal{P}')$ s.t. S' induces connected $(\frac{|S'|}{45}, 2)$ -expander.

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Claim

After Round 2, w.h.p. all of the induced subgraphs $G[S']$ have a Hamilton path.

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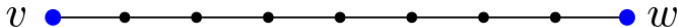
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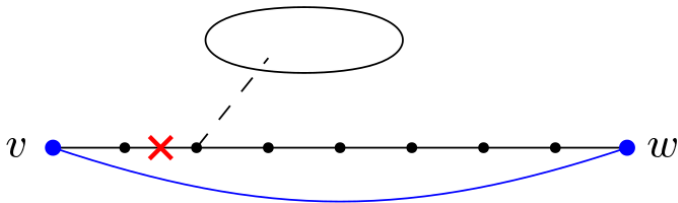
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Conclusions

Let v, w be endpoints of a longest path in H .



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Lemma

If H is connected and has no Hamilton path, then adding the edge $\{v, w\}$ boosts the length of a longest path by ≥ 1 .

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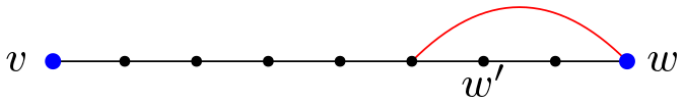
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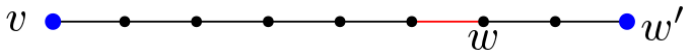
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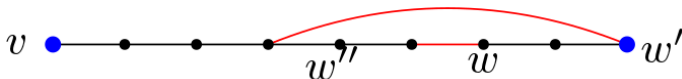
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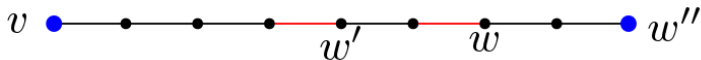
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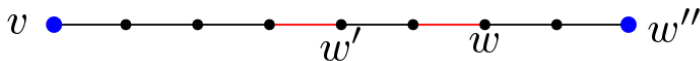
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Definition

$\text{End}(P, v)$ is the set of w which are endpoints of some P' obtained by a sequence of rotations from P with v fixed.



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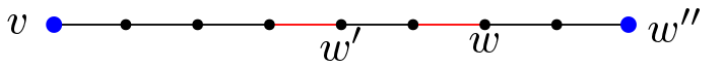
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Lemma (Exercise.)

$$|N(\text{End}(P, v))| \leq 2|\text{End}(P, v)| - 1.$$

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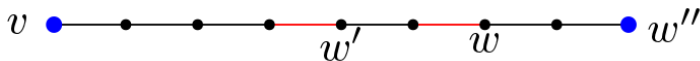
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Corollary

If G is a connected $(r, 2)$ -expander and P a longest path in G , then either P is Hamiltonian or $\exists \geq \frac{r^2}{2}$ potential boosters.

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Conclusions

After round 1

In $G_0 = \mathcal{G}(n, \frac{p}{2})$, each set-pairing $(S, \mathcal{P})$ has sub-pairing $(S', \mathcal{P}')$ s.t. $G_0[S']$ connected and an $(\frac{|S'|}{45}, 2)$ -expander.

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During round 2

S' receives new edges $\{e_1, e_2, \dots, e_r\}$, $r \geq |S'|^2 p/8$ w.h.p.

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We add $\frac{r}{2025} \geq 4n$ boosters to each S' w.h.p., so we're happy.

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The technique also gives $\chi_{\text{lin}}(\mathcal{G}(n, c/n)) = \Theta(n)$ for $c \geq 100000$ (say).

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Theorem (E. & Prałat '23)

W.h.p.,

$$\chi_{\text{lin}}(\mathcal{G}(n, c/n)) = \begin{cases} \Theta(\log \log n) & c \in (0, 1) \\ \Theta(\log n) & c = 1 \end{cases}$$

(Same as χ_{cen} in these regimes.)

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What about $c > 1$ but not too big?

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